Finiteness of Shallow Logical Operators and a Sharp Depth Threshold

The Story Is Still Unfolding*
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For a quantum code on n finite-dimensional physical subsystems with distance d, we consider all exact logical unitaries realized by circuits of depth at most h and circuit gates acting on at most ρ subsystems. For a uniform upper-bound $h < 2\lceil \log_\rho d \rceil - 1$, these circuits can realize only finitely many logical actions up to a global phase, without any stabilizer or geometric locality assumptions. In contrast, for every qubit stabilizer code encoding at least one qubit, a non-trivial continuous one-parameter family of logical unitaries is available at depth $2\lceil \log_\rho d \rceil - 1$.

Controlling the error propagation is the central question in fault-tolerant computation. A single fault in a depth-h circuit with gate range ρ can affect at most ρ^h subsystems. So shallow quantum circuits of bounded range can effectively limit error propagation between error-correction steps. Under the constant-depth and bounded-range premise, Gottesman's threshold construction protects polynomial-size computations below a noise threshold with constant space overhead [1]. Nevertheless, the constant-depth premise seems not sufficient for complete, or even infinitely many logical operations. The celebrated Eastin–Knill theorem [2] shows that transversal gates with respect to a fixed partition into detectable blocks realize only finitely many logical actions up to a global phase. The same restriction holds for constant-depth circuits with geometric locality or algebraic assumptions [3–6]. Therefore, one may ask at what depth this necessary condition can first hold.

As we will show in the following, the precise threshold depth is $2\lceil \log_\rho d \rceil - 1$, given $d, \rho \geq 2$. More specifically, Theorem 1 rules out an infinite logical image when $h < 2\lceil \log_\rho d \rceil - 1$, without any stabilizer or geometric locality assumption. And Theorem 2 shows that every qubit stabilizer code encoding at least one qubit realizes a nontrivial continuous logical family at depth $2\lceil \log_\rho d \rceil - 1$, when arbitrary gates on at most ρ qubits are allowed. Stabilizer subspace codes are thus the most capable candidates for the emergence of an infinite logical family, attaining the smallest possible depth at given d and ρ.

Settings and notations. Let the physical system be $\mathcal{H} = \bigotimes_{i=1}^n \mathcal{H}_i$, with $n \geq 1$ and finite-dimensional factors $\mathcal{H}_i$. The code $\mathcal{C}$ is specified by an isometry $V : \mathcal{H}_L \otimes \mathcal{H}_G \longrightarrow \mathcal{H}$, where $\dim \mathcal{H}_L = K \geq 2$ and $\dim \mathcal{H}_G = g \geq 1$ denote the dimensions of logical and gauge subspaces, respectively. A distance lower bound d means that every operator O supported on fewer than d physical subsystems satisfies

$$V^\dagger O V = I_L \otimes M_O \tag{1}$$

for some operator M_O on $\mathcal{H}_G$ [7, 8]. For unitaries W on $\mathcal{H}_L$ and Z on $\mathcal{H}_G$, set $Q = W \otimes Z$. A physical unitary U implements an exact logical action if such W, Z can be chosen so that

$$UV = VQ = V(W \otimes Z). \tag{2}$$

This equality holds for all logical and gauge states, and the logical action is the phase class $[W] \in \mathrm{PU}(K)$. For integers $h \geq 1$ and $\rho \geq 2$, let $\mathcal{L}_{\leq h}(\mathcal{C}, \rho)$ be the set of these classes obtained when U has at most h layers of arbitrary unitary gates on disjoint supports of at most ρ physical subsystems. The depth bound applies to each complete circuit.

To prove the finiteness, we extend the local rigidity argument of Eastin–Knill [2] and covariant error correction [9, 10] to bounded-depth circuits. For each support arrangement, the circuit equations define a complex affine set containing the physical solutions. The joint evaluation map to $\mathbb{C}^{K^4}$ records the matrix of $X \mapsto WXW^{-1}$ in the matrix-unit basis. We differentiate one gate at a time and use the two light cones in Fig. 1 to compress each insertion to the gauge factor. The derivative of W is then proportional to W, so the conjugation coordinates have zero derivative. The affine set has finitely many irreducible components, corresponding to the minimal primes of the Noetherian coordinate algebra. By Lemma 1, the vanishing of derivatives in each component's function field makes the map send that entire component to a single point. On physical solutions, each output matrix determines $[W]$, since equal conjugation maps imply equality up to phase. Taking the union over the finitely many support arrangements gives a finite logical image.

Theorem 1. *Let $n, h \geq 1$, and $\rho \geq 2$ be integers, and let V encode a quantum code $\mathcal{C}$ satisfying Eq. (1) for a distance $d \geq 2$ on n physical subsystems. If*

$$h < 2\lceil \log_\rho d \rceil - 1, \tag{3}$$

then the logical actions of all circuits of depth at most h, with gates on at most ρ subsystems, that satisfy Eq. (2) obey

$$|\mathcal{L}_{\leq h}(\mathcal{C}, \rho)| < \infty.$$

All gate supports and unitary matrices are allowed. Consequently, $\mathcal{L}_{\leq h}(\mathcal{C}, \rho)$ is not dense in $\mathrm{PU}(K)$.

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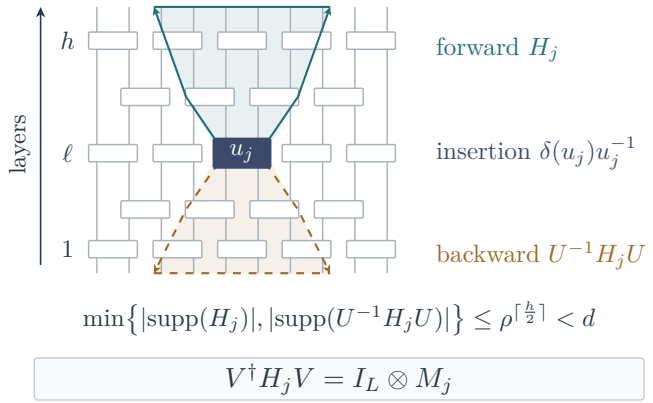


FIG. 1. Two-sided light cones. A derivation applied to gate u_j in layer ℓ gives the propagated insertion H_j. Under the depth condition, its forward support or the backward support of $U^{-1}H_jU$ is detectable. Error detection and the exact encoded action yield gauge-only compression of H_j in the output frame. The brickwall arrangement assumes no geometric locality.

Proof. We pad to h layers and n slots per layer with identity gates on empty supports. Since each slot has only finitely many possible supports, there are finitely many ordered support arrangements α. We fix one and label its $J = nh$ slots in execution order. For a slot with support S_j, take two independent matrices on $\bigotimes_{i\in S_j} \mathcal{H}_i$. Extend both by the identity on the complement and call the resulting matrices u_j, v_j. Write $U = u_J \cdots u_1$. Take a $K\times K$ matrix W and a $g\times g$ matrix Z, and set $Q = W\otimes Z$ as in Eq. (2). The matrices V and $V^\dagger$ are constant, with $V^\dagger V = I_L \otimes I_G$.

Let $\mathbf{x}$ consist of the entries of the local matrices and of W, Z. The gate inverse variables allow us to use polynomial relations. Let $R_\alpha = \mathbb{C}[\mathbf{x}]/I_\alpha$, where I_α is generated by the entrywise differences in

$$u_j v_j = v_j u_j = I, \quad UV = VQ, \quad V^\dagger U = QV^\dagger. \tag{4}$$

The gate relations give $v_j = u_j^{-1}$ and $U^{-1} = v_1 \cdots v_J$. The two encoding relations give

$$(V^\dagger U^{-1} V)Q = V^\dagger U^{-1} UV = I_L \otimes I_G,$$
$$Q(V^\dagger U^{-1} V) = V^\dagger U U^{-1} V = I_L \otimes I_G.$$

Thus $Q^{-1} = V^\dagger U^{-1} V$ over R_α. The factors $W \otimes I_G$ and $I_L \otimes Z$ commute and multiply to Q. Their two-sided inverses are therefore $(I_L \otimes Z)Q^{-1}$ and $(W \otimes I_G)Q^{-1}$, respectively. A diagonal gauge block of the first matrix multiplies W to I_L on either side, so it gives W^{-1}. A diagonal logical block of the second gives Z^{-1} in the same way. Both inverses are matrices over R_α, obtained from the variables already in $\mathbf{x}$.

A physical circuit satisfying Eq. (2) gives a complex evaluation $R_\alpha \to \mathbb{C}$ by assigning the gate adjoints to the v_j. Indeed $UV = VQ$ gives $U^{-1}V = VQ^{-1}$, whose adjoint is $V^\dagger U = QV^\dagger$ because U and Q are unitary. Complex evaluations of the polynomial system also allow nonunitary matrices.

For $1 \le a, b, c, e \le K$, define in R_α

$$r_{abce} = W_{ab}(W^{-1})_{ce}. \tag{5}$$

At a complex evaluation ϕ, write W for the evaluated logical matrix. The conjugation map $X \mapsto WXW^{-1}$ has the expansion

$$W\mathsf{e}_{bc}W^{-1} = \sum_{a,e=1}^{K} \phi(r_{abce})\,\mathsf{e}_{ae},$$

because the (a,e) entry of $W\mathsf{e}_{bc}W^{-1}$ is $W_{ab}(W^{-1})_{ce}$. The joint tuple lists the matrix entries of this map, with $\phi(r_{abce})$ in row (a,e), column (b,c).

Lemma 1 tests derivations in the function fields of the irreducible components. To cover these fields, take an arbitrary extension $F/\mathbb{C}$, a $\mathbb{C}$-algebra map $\phi : R_\alpha \to F$, and a $\mathbb{C}$-linear derivation $\delta : F \to F$. We use the same symbols for the evaluated matrices. The relations above hold over F, with V and $V^\dagger$ still constant. The product rule holds for matrix products when δ is applied entrywise.

The detection identity extends from $\mathbb{C}$ to F in the chosen tensor-product bases. If O is supported on a set of fewer than d subsystems, write $O = \sum_\mu a_\mu B_\mu$, with $a_\mu \in F$ and B_μ the complex matrix units on that set extended by identities. Each B_μ satisfies Eq. (1). Taking the same F-linear combination gives $V^\dagger O V = I_L \otimes \sum_\mu a_\mu M_{B_\mu}$.

Put $U_{>j} = u_J \cdots u_{j+1}$ and $U_{<j} = u_{j-1} \cdots u_1$, with an empty product equal to the identity. The product rule gives

$$\delta(U) = \sum_{j=1}^{J} U_{>j}\,\delta(u_j)\,U_{<j}.$$

In the jth term of $\delta(U)U^{-1}$, write $U^{-1} = U_{<j}^{-1} u_j^{-1} U_{>j}^{-1}$. The adjacent factors $U_{<j}U_{<j}^{-1}$ cancel, giving

$$H = \delta(U)U^{-1} = \sum_{j=1}^{J} U_{>j}\,\delta(u_j)u_j^{-1}U_{>j}^{-1} = \sum_{j=1}^{J} H_j. \tag{6}$$

In the input frame, substituting $U = U_{>j}u_jU_{<j}$ cancels the factors $U_{>j}$ and $U_{>j}^{-1}$, leaving

$$\begin{aligned} U^{-1}H_jU &= U_{<j}^{-1}u_j^{-1}\big(\delta(u_j)u_j^{-1}\big)u_jU_{<j} \\ &= U_{<j}^{-1}u_j^{-1}\delta(u_j)U_{<j}. \end{aligned}$$

The complementary identity factors are constant, so $\delta(u_j)$ and both local products $\delta(u_j)u_j^{-1}$ and $u_j^{-1}\delta(u_j)$ are supported on S_j. The other gates in the same layer are disjoint from S_j and leave each product unchanged by conjugation. In any other layer, include single-site identities on the untouched sites. Conjugating an operator supported on T can spread it only to the union of the

disjoint gate blocks that meet T. At most $|T|$ blocks meet T, and each has size at most ρ. The same bound applies to conjugation by the inverse layer.

For slot j in layer ℓ, the initial support has size at most ρ, with $h-\ell$ layers to propagate through toward the output and $\ell-1$ toward the input. The two support bounds are therefore $\rho^{h-\ell+1}$ and ρ^{ℓ} (Fig. 1). The smaller is at most $\rho^{\lceil h/2\rceil}$ because $\min\{h-\ell+1,\ell\} \le \lceil h/2\rceil$. For $m=\lceil \log_\rho d\rceil$, the integer depth in Eq. (3) satisfies $h \le 2m-2$. Hence the smaller support has size at most $\rho^{m-1} < d$.

If the output support is detectable, Eq. (1) gives $V^\dagger H_j V = I_L \otimes M_j$ directly. Otherwise detection in the input frame gives $V^\dagger U^{-1} H_j U V = I_L \otimes N_j$. Multiplying $V^\dagger U = Q V^\dagger$ on the left by Q^{-1} and on the right by U^{-1} gives $V^\dagger U^{-1} = Q^{-1} V^\dagger$. With $UV = VQ$, the input compression is therefore related to the output compression by

$$V^\dagger U^{-1} H_j U V = Q^{-1}(V^\dagger H_j V)Q,$$
$$V^\dagger H_j V = Q(I_L \otimes N_j)Q^{-1} = I_L \otimes Z N_j Z^{-1}.$$

Thus in either case $V^\dagger H_j V = I_L \otimes M_j$ in the output frame. Summing over j, we have

$$V^\dagger H V = I_L \otimes M, \qquad M = \sum_{j=1}^{J} M_j. \tag{7}$$

Since V and $V^\dagger$ are constant, differentiating $UV = VQ$ gives $\delta(U)V = V\delta(Q)$. With $U^{-1}V = VQ^{-1}$, this yields

$$V^\dagger H V = V^\dagger \delta(U) V Q^{-1} = V^\dagger V \delta(Q) Q^{-1} = \delta(Q)Q^{-1}.$$

The product rule for $Q = W \otimes Z$, followed by right multiplication by $Q^{-1} = W^{-1} \otimes Z^{-1}$, gives

$$\delta(Q)Q^{-1} = \delta(W)W^{-1} \otimes I_G + I_L \otimes \delta(Z)Z^{-1}. \tag{8}$$

By Eq. (7), the left side is $I_L \otimes M$. Subtracting the gauge term and taking the $(1,1)$ gauge block gives $\delta(W)W^{-1} = \lambda I_L$, with $\lambda = (M - \delta(Z)Z^{-1})_{11} \in F$.

Thus $\delta(W) = \lambda W$, and differentiating $WW^{-1} = I_L$ gives

$$\delta(W^{-1}) = -W^{-1}\delta(W)W^{-1} = -\lambda W^{-1}.$$

The product rule for $\phi(r_{abce}) = W_{ab}(W^{-1})_{ce}$ then gives

$$\delta\big(\phi(r_{abce})\big) = \lambda\phi(r_{abce}) - \lambda\phi(r_{abce}) = 0.$$

Apply Lemma 1 to the finitely generated algebra R_α and the K^4 coordinates in Eq. (5). For each minimal prime $\mathfrak{p}$, take $F = F_\mathfrak{p} = \mathrm{Frac}(R_\alpha/\mathfrak{p})$ and $\phi = \iota_\mathfrak{p}$. Our calculation for arbitrary δ verifies the vanishing hypothesis in each of these function fields.

In the lemma's proof, a transcendental coordinate would admit a derivation sending it to 1, so the vanishing forces one constant tuple per component. Every complex evaluation has a kernel containing a minimal prime and therefore takes one of those finitely many tuples. By the matrix-unit expansion, the corresponding conjugation matrices include all those of the physical circuits for the chosen support arrangement.

If two physical evaluations give the same conjugation matrix, then $W_1 X W_1^{-1} = W_2 X W_2^{-1}$ for every logical matrix X. Multiplying on the left by W_2^{-1} and on the right by W_1 gives

$$(W_2^{-1}W_1)X = X(W_2^{-1}W_1).$$

Commutation with the diagonal matrix units makes $W_2^{-1}W_1$ diagonal, and commutation with the off-diagonal units makes its diagonal entries equal. By unitarity, $W_1 = e^{i\theta}W_2$. A scalar phase cancels in $X \mapsto WXW^{-1}$, so each realized conjugation matrix gives exactly one class $[W]$. The finite union over support arrangements therefore gives a finite set of classes, which is closed and proper in the infinite Hausdorff space $\mathrm{PU}(K)$ for $K \ge 2$. □

Finiteness does not exclude a finite subset $S \subseteq \mathcal{L}_{\le h}(\mathcal{C},\rho)$ for which

$$\overline{\langle S\rangle} = \mathrm{PU}(K).$$

The group $\langle S\rangle$ allows words of arbitrary finite length. A word of length r in $S \cup S^{-1}$ has a concatenated implementation of depth at most rh. These products are not required to remain in $\mathcal{L}_{\le h}(\mathcal{C},\rho)$, so density of the generated group does not assert density within the original depth bound. For example, Dua et al. give a $[[37,1,7]]$ CSS code with separate depth-one circuits of gate range $\rho = 3$ that implement [11, Sec. V and Appendix B 3]

$$S = \{\overline{T}^\dagger, \sqrt{\overline{X}}^{\,\dagger}\} \subseteq \mathcal{L}_{\le 1}(\mathcal{C},3).$$

The two circuits use different partitions of the physical qubits. Since $\overline{H} \propto (\overline{T}^\dagger)^2 \sqrt{\overline{X}}^{\,\dagger} (\overline{T}^\dagger)^2$ and $\overline{T} = (\overline{T}^\dagger)^7$, this pair generates a dense subgroup of PU(2). For this code, $2\lceil \log_3 7\rceil - 1 = 3$, so $\mathcal{L}_{\le 1}(\mathcal{C},3)$ is finite by Theorem 1.

In a fault-tolerance threshold construction based on the constant-depth unitary logical gadgets considered here, logical-gate compilation is necessary for universal control in the finite-image regime. On a fixed code, the available shallow logical actions form a finite alphabet. If its generated group is dense, arbitrary target logical unitaries can be approximated by words over this alphabet, with error correction interleaved as prescribed by the protocol [1, 11]. The constant-depth premise applies to each gadget. Compiled word lengths cannot remain uniformly bounded when arbitrary targets must be approximated to arbitrarily high accuracy.

The finiteness condition has an exact boundary for stabilizer codes. For an integer $\rho \ge 2$, define the critical depth

$$h_\infty(\mathcal{C},\rho) = \min\{h \in \mathbb{Z}_{\ge 1} : \mathcal{L}_{\le h}(\mathcal{C},\rho) \text{ is infinite}\}, \tag{9}$$

with value ∞ if the set is empty. When $d \geq 2$, Theorem 1 gives

$$h_\infty(\mathcal{C}, \rho) \geq 2\lceil \log_\rho d \rceil - 1. \tag{10}$$

For the upper bound of $h_\infty(\mathcal{C}, \rho)$, we implement continuous rotations generated by a weight-d logical Pauli. A Clifford tree collects its nonidentity factors onto at most ρ qubits in $\lceil \log_\rho d \rceil - 1$ layers. The compressed Pauli then fits in one allowed rotation gate. Conjugating back recovers the original Pauli rotation, at total depth $2\lceil \log_\rho d \rceil - 1$. This uses the established tree constructions for parity compression, phase gadgets, and Pauli-rotation synthesis [12–14]. The resulting continuous family is nonconstant modulo phase, giving the matching upper bound.

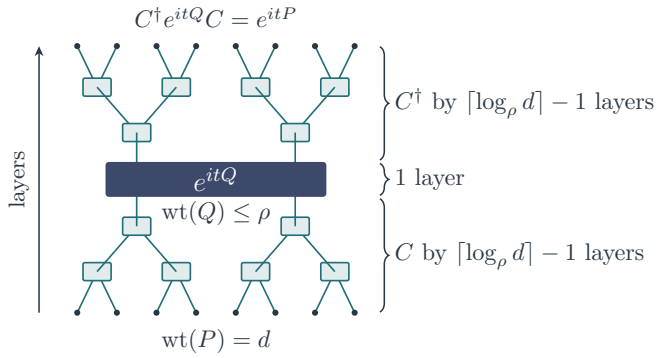


FIG. 2. Mechanism of Theorem 2. Time runs upward. C compresses the support of a weight-d logical Pauli to $Q = CPC^\dagger$, with $\mathrm{wt}(Q) \leq \rho$. One rotation gate followed by $C^\dagger$ gives the complete logical circuit $C^\dagger e^{itQ} C = e^{itP}$, of depth $2\lceil \log_\rho d \rceil - 1$.

Theorem 2. *Let $\mathcal{C}$ be an $[[n, k, d]]$ qubit stabilizer code, with $k \geq 1$ and actual distance $d \geq 2$. Let $\rho \geq 2$ be an integer, and allow arbitrary unitary gates on any set of at most ρ qubits. Then*

$$h_\infty(\mathcal{C}, \rho) = 2\lceil \log_\rho d \rceil - 1. \tag{11}$$

At this depth, the logical image contains a nontrivial continuous one-parameter subgroup.

Proof. Put $m = \lceil \log_\rho d \rceil$. It suffices to construct infinitely many logical actions at depth at most $2m - 1$. Choose a Hermitian code-preserving Pauli P of weight d, with $P^2 = I$ and nonscalar logical action $\overline{P}$ [15, Sec. 3.2]. We first compress its support so that its rotation fits in one allowed gate.

We adapt the unitary parity-concentration part of the binary-tree in-line symmetry-verification circuit in Ref. [12, Sec. IV and Fig. 2(a)] to gates on at most ρ qubits. At each layer, partition the active factors into groups of ρ, with at most one smaller group. In each group, basis changes and CNOTs compose into one allowed Clifford gate collecting the product onto one qubit. The disjoint groups form one layer. If w_r is the weight after r layers, then $w_0 = d$ and $w_{r+1} = \lceil \frac{w_r}{\rho} \rceil$, giving $w_r = \lceil \frac{d}{\rho^r} \rceil$. Thus $m - 1$ layers yield C with $Q = CPC^\dagger$ of weight at most ρ. The compressed rotation e^{itQ} is one allowed gate. Conjugating back gives

$$U(t) = C^\dagger e^{itQ} C = e^{itP}. \tag{12}$$

The depth is at most $(m - 1) + 1 + (m - 1) = 2m - 1$. The complete circuit preserves the code and induces $e^{it\overline{P}}$. As $\overline{P}$ has both eigenvalues ± 1, its projective rotations form a nontrivial continuous U(1) subgroup of period π. Hence $h_\infty(\mathcal{C}, \rho) \leq 2m - 1$, and the lower bound gives Eq. (11). □

The construction attains the threshold for an infinite exact logical image, but does not by itself provide fault tolerance against arbitrary single-gate faults. An error $e^{i\varepsilon Q}$ immediately after the central rotation gate propagates through $C^\dagger$ to $e^{i\varepsilon P}$, which induces a nontrivial logical error for $\varepsilon \notin \pi\mathbb{Z}$.

Conclusion. For general quantum codes in the finite-dimensional setting considered here, bounded-range circuits below the stated threshold realize only finitely many exact logical actions up to phase. The proof combines a detectable light cone in one of the two circuit directions with a finite-image criterion for the circuit coordinate algebra. Qubit stabilizer codes attain the bound by concentrating the support of a logical Pauli into one rotation gate and then reversing the compression. Together, the theorems locate the critical circuit depth to support infinitely many exact logical actions without imposing geometric locality.

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Lean proofs of the statements of Theorems 1 and 2, together with verification scripts, are available in the project repository (proof snapshot `ad6a378`). The proof of Theorem 1 follows the coordinate-algebra and derivation argument presented here, including scalar extension of detection, two-sided support bounds and the finite-evaluation-image criterion. The audited theorem dependencies contain no project axioms or `sorryAx`; their foundational axioms are `propext`, `Classical.choice` and `Quot.sound`. The quantitative bounds and computational discussion in the End Matter are additional mathematical deductions and are not formalized in this snapshot.

[1] D. Gottesman, Quantum Inf. Comput. **14**, 1338–1371 (2014), arXiv:1310.2984.
[2] B. Eastin and E. Knill, Phys. Rev. Lett. **102**, 110502 (2009), arXiv:0811.4262.

[3] S. Bravyi and R. König, Phys. Rev. Lett. **110**, 170503 (2013), arXiv:1206.1609.
[4] F. Pastawski and B. Yoshida, Phys. Rev. A **91**, 012305 (2015), arXiv:1408.1720.
[5] T. Jochym-O'Connor, A. Kubica, and T. J. Yoder, Phys. Rev. X **8**, 021047 (2018), arXiv:1710.07256.
[6] P. Webster, M. Vasmer, T. R. Scruby, and S. D. Bartlett, Phys. Rev. Research **4**, 013092 (2022), arXiv:2012.05260.
[7] E. Knill and R. Laflamme, Phys. Rev. A **55**, 900 (1997), arXiv:quant-ph/9604034.
[8] D. Kribs, R. Laflamme, and D. Poulin, Phys. Rev. Lett. **94**, 180501 (2005), arXiv:quant-ph/0412076.
[9] P. Faist, S. Nezami, V. V. Albert, G. Salton, F. Pastawski, P. Hayden, and J. Preskill, Phys. Rev. X **10**, 041018 (2020), arXiv:1902.07714.
[10] Z.-W. Liu and S. Zhou, npj Quantum Inf. **9**, 119 (2023), arXiv:2111.06355. See also its long supplement, arXiv:2111.06360, Sec. VI.
[11] A. Dua, A. Holmes, M. J. Gullans, and V. V. Albert, *Qubit CSS codes with Universal Transversal Gates*, arXiv:2610.06730v1 (2026).
[12] X. Bonet-Monroig, R. Sagastizabal, M. Singh, and T. E. O'Brien, Phys. Rev. A **98**, 062339 (2018), arXiv:1807.10050.
[13] A. Cowtan, S. Dilkes, R. Duncan, W. Simmons, and S. Sivarajah, Electron. Proc. Theor. Comput. Sci. **318**, 213 (2020), arXiv:1906.01734.
[14] J. Haah, Y. Liu, and X. Tan, Commun. Math. Phys. **406**, 309 (2025), arXiv:2402.05239.
[15] D. Gottesman, *Stabilizer Codes and Quantum Error Correction*, Ph.D. thesis, California Institute of Technology (1997), arXiv:quant-ph/9705052.
[16] The Stacks Project Authors, Lemma 10.31.6 (Tag 00FR) and Lemma 10.17.2 (Tag 00E0), *The Stacks Project*.
[17] P. Bürgisser and P. Scheiblechner, *On the complexity of counting components of algebraic varieties*, J. Symbolic Comput. **44**, 1114–1136 (2009); author manuscript, Secs. 2.1.2–2.1.3.
[18] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed. (Springer, Berlin, 2006).

END MATTER

Algebraic finite-image criterion

We give the global finiteness step in detail. The objects of interest are finitely many elements $r_1, \dots, r_m$ of a coordinate algebra R, representing the output coordinates of a polynomial map. An evaluation is a unital $\mathbb{C}$-algebra homomorphism $\phi : R \to \mathbb{C}$. We seek a finite set containing all output tuples $(\phi(r_1), \dots, \phi(r_m))$; neither the set of evaluations nor the whole parameter space is required to be finite.

For a minimal prime ideal $\mathfrak{p}$ of R, put

$$F_{\mathfrak{p}} = \mathrm{Frac}(R/\mathfrak{p}), \qquad \iota_{\mathfrak{p}} : R \longrightarrow R/\mathfrak{p} \longrightarrow F_{\mathfrak{p}}.$$

The quotient is an integral domain, so its map into its fraction field is injective and $\ker \iota_{\mathfrak{p}} = \mathfrak{p}$. A $\mathbb{C}$-linear derivation $\delta : F_{\mathfrak{p}} \to F_{\mathfrak{p}}$ is a linear map satisfying $\delta(xy) = \delta(x)y + x\delta(y)$. The derivations must be taken in these function fields: a $\mathbb{C}$-linear derivation of $\mathbb{C}$ itself is always zero and would give no constraint on the outputs.

Lemma 1. *Let R be a finitely generated commutative $\mathbb{C}$-algebra, and let $r_1, \dots, r_m \in R$. Suppose that, for every minimal prime $\mathfrak{p}$ and every $\mathbb{C}$-linear derivation $\delta : F_{\mathfrak{p}} \to F_{\mathfrak{p}}$,*

$$\delta\big(\iota_{\mathfrak{p}}(r_i)\big) = 0 \quad \textit{for every } i.$$

Then the joint evaluation image

$$\{(\phi(r_1), \dots, \phi(r_m)) : \phi : R \to \mathbb{C} \ \textit{a } \mathbb{C}\textit{-algebra map}\}$$

is finite.

Proof. If $R = 0$, there is no unital map $R \to \mathbb{C}$, and the image is empty. Assume $R \neq 0$.

Constants in a function field. Fix a minimal prime $\mathfrak{p}$ and write $F = F_{\mathfrak{p}}$. The images of the finitely many algebra generators of R generate F as a field over $\mathbb{C}$. Suppose $a \in F$ is killed by every $\mathbb{C}$-linear derivation. If a were transcendental, we could choose a transcendence basis B of $F/\mathbb{C}$ containing a. On the rational function field $L = \mathbb{C}(B)$, let ∂ be partial differentiation with respect to a. Then $\partial(a) = 1$, with the other elements of B and all complex numbers held constant.

By the choice of B, the field F is algebraic over L. It is generated over L by finitely many algebraic elements, so F/L is finite. It is separable in characteristic zero, and the primitive element theorem gives $F = L(\beta)$. Let $f(T) = T^s + \sum_{j=0}^{s-1} a_j T^j$ be the minimal polynomial of β. Differentiating $f(\beta) = 0$ prescribes the value

$$c = -\frac{\sum_{j=0}^{s-1} \partial(a_j)\beta^j}{f'(\beta)}$$

for the derivative of β. The denominator is nonzero by separability. To construct the extension, for $g(T) = \sum_j b_j T^j \in L[T]$ set

$$D(g) = \sum_j \partial(b_j)\beta^j + g'(\beta)c.$$

The product rule gives $D(gh) = D(g)h(\beta) + g(\beta)D(h)$. The choice of c makes $D(f) = 0$, and $f(\beta) = 0$, so $D(fh) = 0$ for every h. Consequently D descends through $L[T]/(f) \cong F$ to a derivation on F extending ∂. It sends a to 1, contrary to the assumption. Thus a is algebraic over $\mathbb{C}$, and algebraic closedness gives $a \in \mathbb{C}$.

By the vanishing hypothesis, this applies to every $\iota_{\mathfrak{p}}(r_i)$. Write $\iota_{\mathfrak{p}}(r_i) = c_{i,\mathfrak{p}} \in \mathbb{C}$. The kernel calculation above therefore gives

$$r_i - c_{i,\mathfrak{p}} \in \mathfrak{p}. \tag{13}$$

From component constants to all evaluations. Hilbert's basis theorem makes R Noetherian. Its minimal primes

form a finite nonempty set $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_t\}$ [16]. For any complex evaluation ϕ, the kernel is prime and contains some $\mathfrak{p}_\nu$ [16]. Applying ϕ to Eq. (13) gives $\phi(r_i) = c_{i,\mathfrak{p}_\nu}$ for every i, with the same ν for the whole tuple. If the evaluation lies on several components, each of them gives that same tuple at this point. Thus at most t joint tuples occur, which proves finiteness.

Monic identities in the coordinate algebra. The same constants yield the monic polynomial identities in R used for the integrality step of the formal proof. For each output coordinate form the polynomial

$$P_i(T) = \prod_{\nu=1}^{t} (T - c_{i,\mathfrak{p}_\nu}).$$

For any ν, one factor of $P_i(r_i)$ lies in $\mathfrak{p}_\nu$ by Eq. (13). Consequently

$$P_i(r_i) \in \bigcap_{\nu=1}^{t} \mathfrak{p}_\nu = \sqrt{(0)}.$$

Every prime contains a minimal prime, so the intersection of the minimal primes equals the intersection of all primes, namely the nilradical [16]. Thus $P_i(r_i)$ is nilpotent. Taking a positive power gives $P_i(r_i)^{a_i} = 0$, so r_i satisfies the monic polynomial $P_i(T)^{a_i}$ in R.

At any complex evaluation ϕ, this identity becomes

$$0 = \phi\big(P_i(r_i)^{a_i}\big) = P_i\big(\phi(r_i)\big)^{a_i}.$$

Since a field has no nonzero nilpotents, $P_i(\phi(r_i)) = 0$. These equations place the joint image in the product of the finite coordinate root sets. The component argument also records which coordinate values occur together, using a single minimal prime for the entire tuple. □

Geometrically, the minimal primes describe the irreducible components of the affine parameter space. The field-derivation calculation makes each output coordinate constant on every such component, and Noetherianity limits their number. The polynomial argument also handles nilpotent directions: for example, in $R = \mathbb{C}[t, \epsilon]/(\epsilon^2)$, the element $r = \epsilon$ is nonzero but every complex evaluation sends it to zero. The parameter t remains arbitrary. Thus a finite output image does not imply that the coordinate algebra is reduced or that its spectrum is zero-dimensional. No smoothness assumption or restriction to regular points enters the argument.

In Theorem 1, $R = R_\alpha$ and the outputs are the K^4 elements in Eq. (5). The calculation for arbitrary F, ϕ, δ includes $F = F_\mathfrak{p}$ and $\phi = \iota_\mathfrak{p}$, as required by the lemma. Its conclusion applies to all complex evaluations, including physical evaluations at singular points. At a physical evaluation $W^{-1} = W^\dagger$, and the tuple is the matrix of $X \mapsto WXW^\dagger$. The scalar-commutant argument in the theorem's proof identifies each realized matrix with a unique phase class $[W]$. The cited Lean snapshot uses the monic identities above and includes variables for the logical and gauge inverses. The block formulas in Theorem 1 recover those entries from the generators of R_α.

Quantitative parameters and exact counting

The same component constants yield a cardinality bound. The kernel of any evaluation $R_\alpha \to \mathbb{C}$ contains a minimal prime $\mathfrak{p}$. All its logical conjugation coordinates therefore have the single joint value $(c_{i,\mathfrak{p}})_i$. Hence

$$|\mathcal{L}_{\le h}(\mathcal{C}, \rho)| \le \sum_\alpha \#\operatorname{Min}(R_\alpha). \tag{14}$$

Different components can give the same logical action, and some algebraic solutions need not be physical, so equality is not asserted.

Here are explicit parameters for the presentation in the proof. Write $q_i = \dim \mathcal{H}_i$, $P = \prod_i q_i$, and $D_j = \prod_{i \in S_j} q_i$, taking $D_j = 1$ on empty supports. With $J = nh$ slots, the number of complex variables is

$$N_\alpha = 2 \sum_{j=1}^{J} D_j^2 + K^2 + g^2. \tag{15}$$

There are at most $2JP^2 + 2PKg$ scalar defining relations, each of degree at most $\Delta = \max\{2, J\}$. The term $2JP^2$ counts the two inverse relations for each embedded physical gate; it is not an optimized presentation. The affine Bézout bound, counting irreducible components of all dimensions, gives $\#\operatorname{Min}(R_\alpha) \le \Delta^{N_\alpha}$ [17]. Nilpotents do not add components or complex evaluations.

A support arrangement assigns each site in each layer either to one of the n labelled slots or to no slot. There are therefore at most $(n+1)^{nh}$ arrangements; disjointness and the arity restriction only reduce this count. Set $q_{\max} = \max_i q_i$ and $s = \min\{\rho, n\}$. Since $D_j \le q_{\max}^s$, under the hypotheses of Theorem 1 we obtain

$$\begin{aligned} |\mathcal{L}_{\le h}(\mathcal{C}, \rho)| &\le (n+1)^{nh} \Delta^{N_*}, \\ N_* = 2nh\, q_{\max}^{2s} + K^2 + g^2, &\qquad \Delta = \max\{2, nh\}. \end{aligned} \tag{16}$$

This bound is uniform over encodings with these dimensions, but very loose. For $n = 5$, $h = \rho = 2$, $q_{\max} = K = 2$, and $g = 1$, it is $6^{10}10^{325}$. A bound depending only on d, ρ, h cannot hold across all codes: the k-fold tensor power of the $[[5, 1, 3]]$ stabilizer code keeps $d = 3, \rho = 2, h = 1$, while its depth-one logical image contains at least 4^k projective logical Paulis.

Exact counting requires more than a component bound. If the encoding entries are supplied as exact algebraic numbers, introduce variables y_i for the logical conjugation coordinates. Use the polynomial representatives of r_i given by Eq. (5) and the gauge block of $(I_L \otimes Z)V^\dagger(v_1 \cdots v_J)V$ for W^{-1}. Eliminate the original matrix-entry variables

from $I_\alpha + (y_i - r_i)_i$. The joint image is finite and therefore Zariski closed, so the elimination ideal describes exactly the complex output tuples. For each candidate tuple, substitute the gate adjoints for the v_j in the coordinate equalities. Test those equalities together with the unitary and encoding equations in real and imaginary parts. Duplicate actions from different components and arrangements must also be identified. Real quantifier elimination provides a decision procedure for this feasibility step [18]. The inverse variables of the enlarged complex model alone do not enforce physical unitarity.

These procedures are effective in principle with exact algebraic input, but the variable count and elimination costs make direct enumeration impractical in general. Arbitrary unspecified complex encoding coefficients do not supply an effective input representation, even though the theorem and Eq. (16) still hold. Exploiting symmetries, smaller presentations or a particular code may improve the calculation. Equations (14)–(16), the Bézout estimate and the exact-counting procedure are not formalized or executed by the cited Lean development.